Lecture 3: Expectation and Joint Distributions

Math for Data Science · 2026

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Agenda

- Part 0: Recap
- Part I: Uniform distribution and CDF
- Part II: Functions of Random Variables
- Part III: Independence of Random Variables
- Part IV: Expectation
- Part V: Joint Distributions

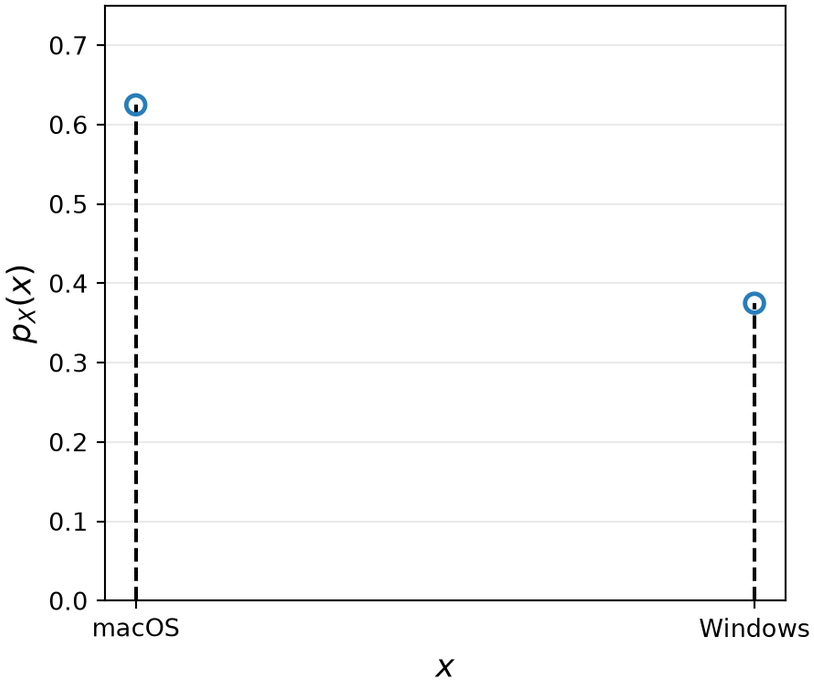
Part 0: Recap

Random Variables

Definition: Given an experiment with sample space S, a random variable is a function from the sample space S to the real numbers $\mathbb{R}$.

Thus, a random variable X assigns a numerical value $X(s)$ to each possible outcome s of the experiment.

- Nonnegative: $p_X(x) > 0$ if $x = x_j$ for some j, and $p_X(x) = 0$ otherwise;
- Sums to 1: $\sum_{j=1}^{\infty} p_X(x_j) = 1$



Part I: Uniform distribution and CDF

The Discrete Uniform Distribution

Let C be a finite, nonempty set of numbers. Choose one of these numbers uniformly at random. Call the chosen number X. Then X is said to have the Discrete Uniform distribution with parameter C.

The PMF of $X \sim \mathrm{DUnif}(C)$ is:

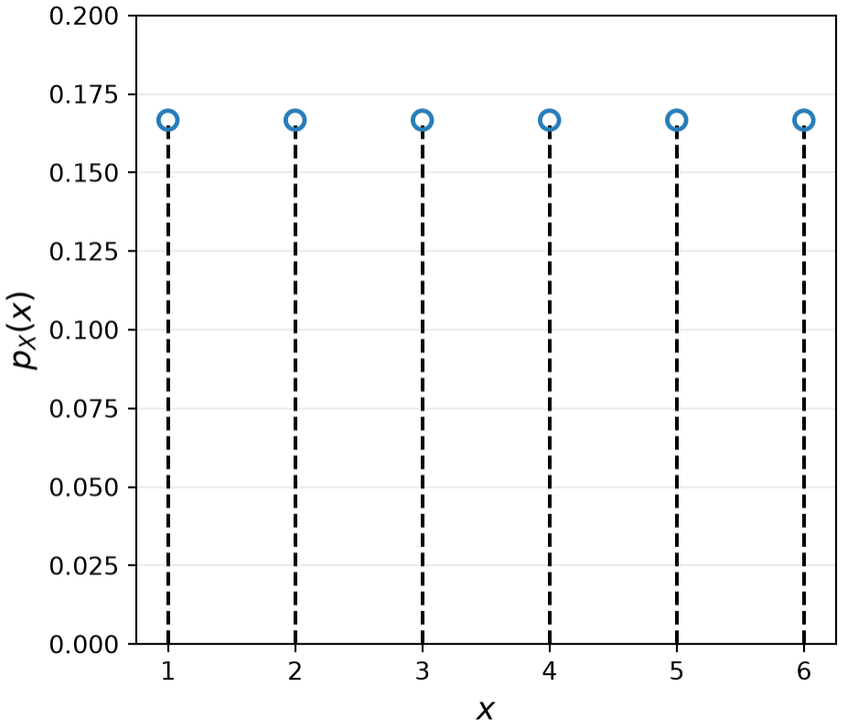
$$P(X = x) = \frac{1}{|C|}$$

where $|C|$ represents the number of elements in the set C.

Similarly,

$$P(X \in A) = \frac{|A|}{|C|}$$

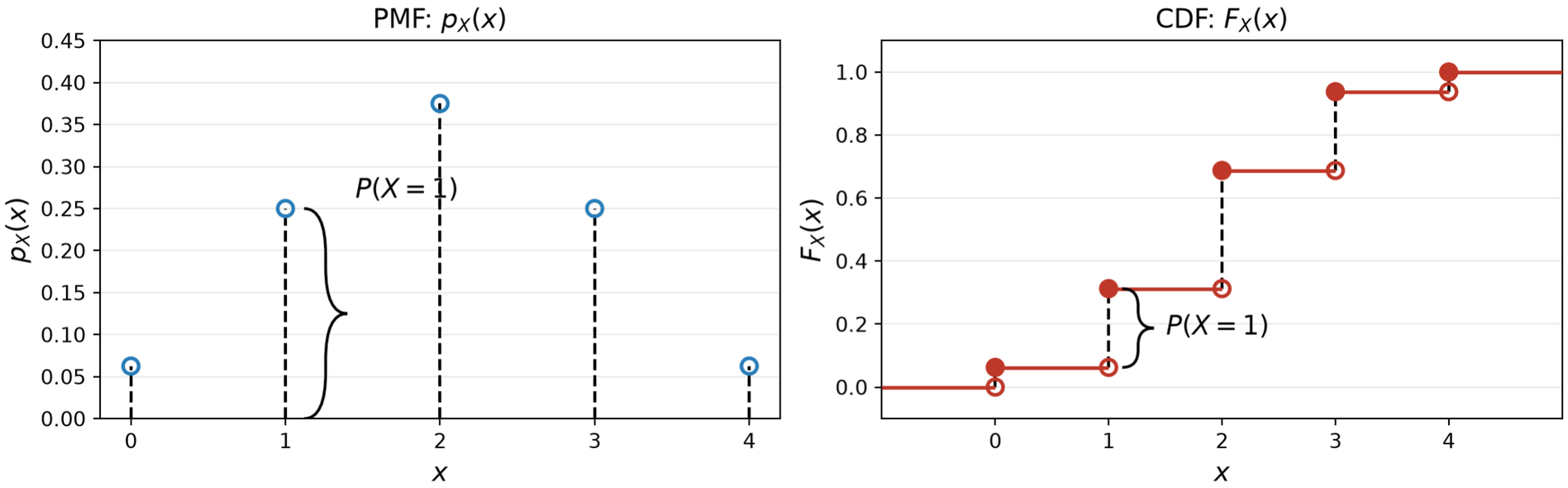
where $A \subseteq C$.



The Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) of a random variable X is the function $F_X(x) = P(X \leq x)$.

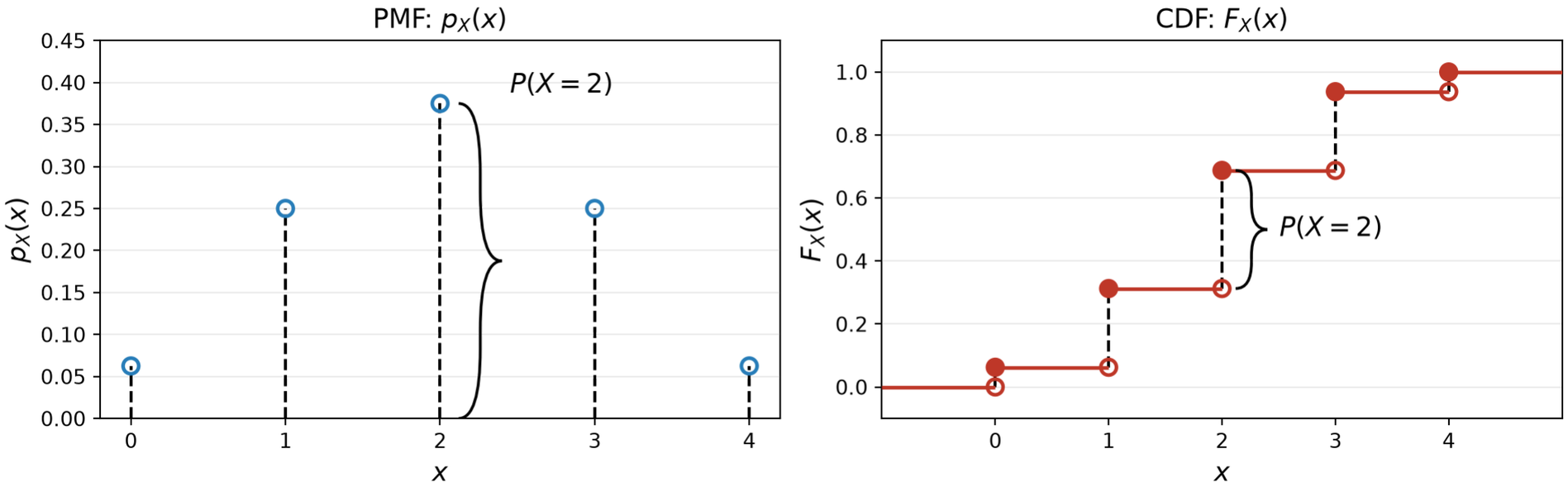
Example: Let $X \sim \square(4, 1/2)$. Then the PMF and CDF of X are given by:



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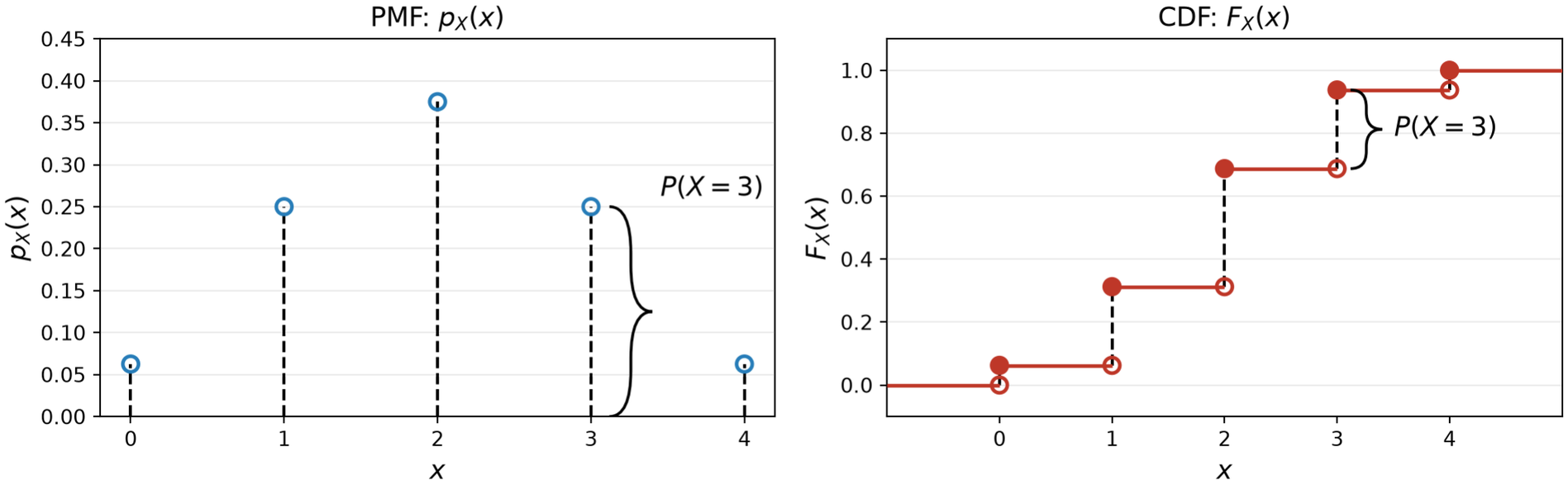
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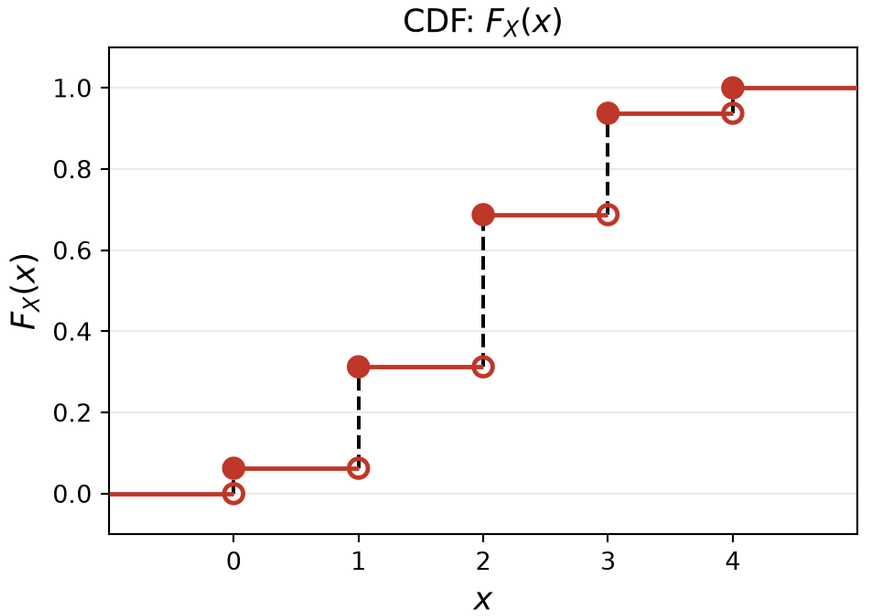


Properties of the CDF

The cumulative distribution function (CDF) of a random variable X is the function $F_X(x) = P(X \leq x)$.

Properties

- valid for discrete and continuous random variables (R.V.)!
- increasing
- right continuous
- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$
- gets the probability of an intervall easily, e.g. $P(x_1 \leq X \leq x_2) = F_X(x_2) - F_X(x_1)$.



Part II: Functions of Random Variables

Functions of a Single Random Variable

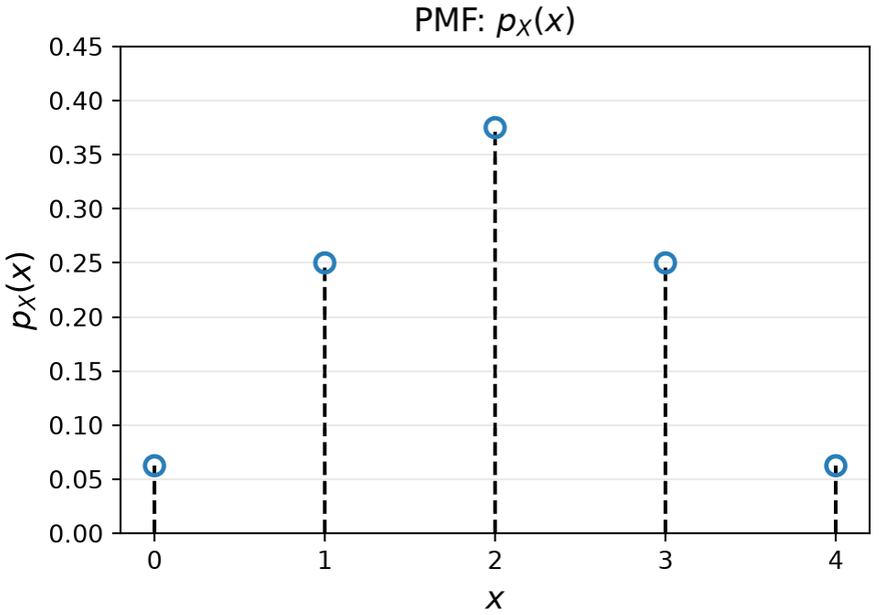
Definition: For an experiment with sample space S, a R.V. X, and a one-to-one function g, $g(X)$ is the random variable that maps s to $g\big(X(s)\big)$ for all $s \in S$.

In other words:

1. First apply X
2. Then take $g(X)$

Assuming g is one-to-one,

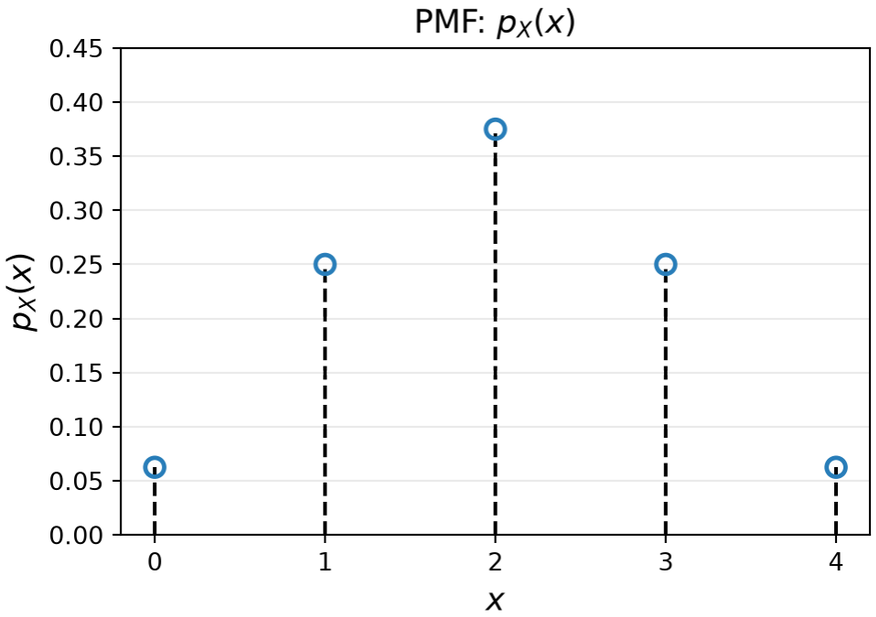
$$P\Big(Y = g(x)\Big) = P(X = x)$$

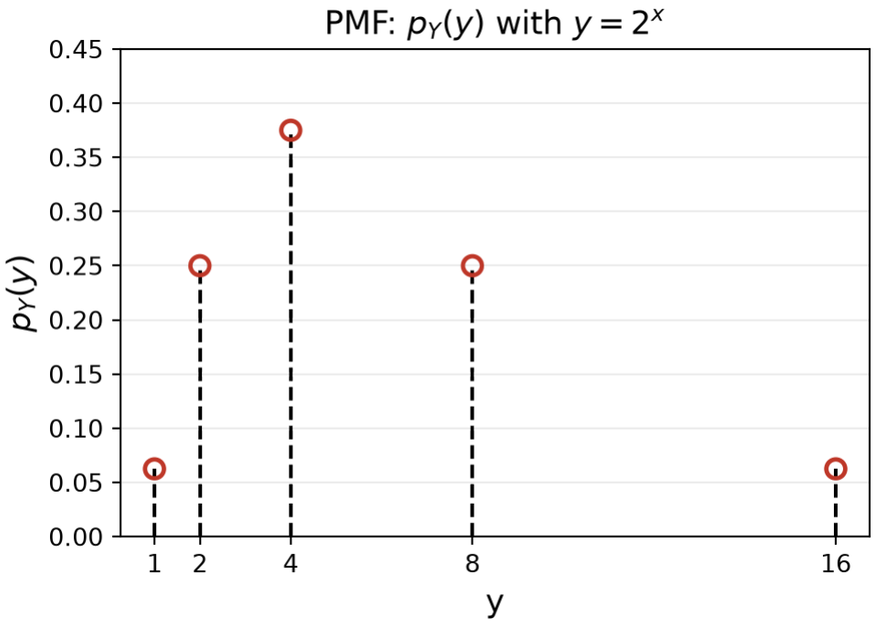


Functions of a Single Random Variable

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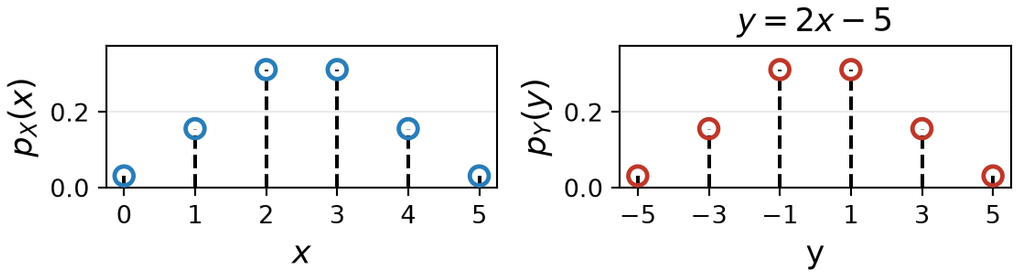
ⓘ **The probability travels with the value. Only its label changes.**

Example: A Random Walk

Example 3.7.2: A person stumbling home moves 1 step to the left or 1 step to the right with equal probability. All steps are independent. Let Y be their position after n steps. What is the PMF of Y?

- Consider every step a Bernoulli trial, with *right* a success and *left* a failure.
- The number of steps to the right after n steps is then $X \sim \text{Bin}(n, 1/2)$.
- If $X = j$, the number of steps to the left is $n - j$, so the final position is $j - (n - j) = 2j - n$.
- Define the final position as the R.V. $Y = 2X - n$:

$$P(Y = y) = P(2X - n = y) = P\left(X = \tfrac{n+y}{2}\right) = \binom{n}{\frac{n+y}{2}}\left(\frac{1}{2}\right)^n$$



A Function of Two Random Variables

Definition: Given an experiment with sample space S, if X and Y are R.V.s that map $s \in S$ to $X(s)$ and $Y(s)$, then $g(X, Y)$ is the R.V. that maps s to $g\Big(X(s), Y(s)\Big)$.

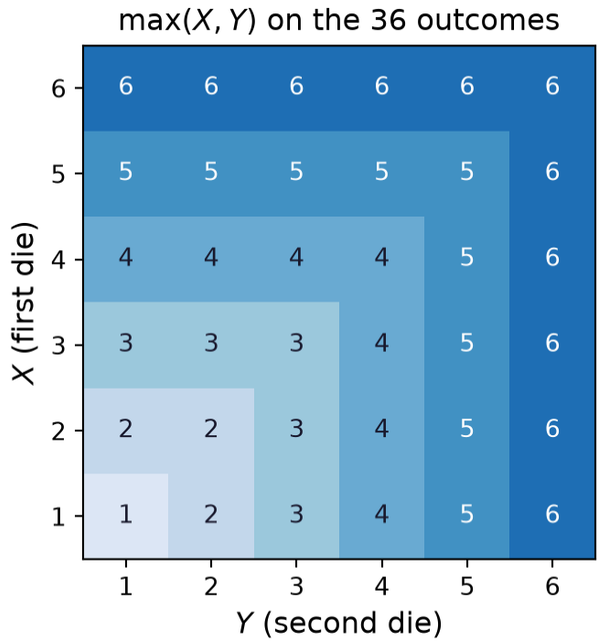
So $X + Y$, XY, and $\max(X, Y)$ are all random variables - each one is just another function on the *same* sample space.

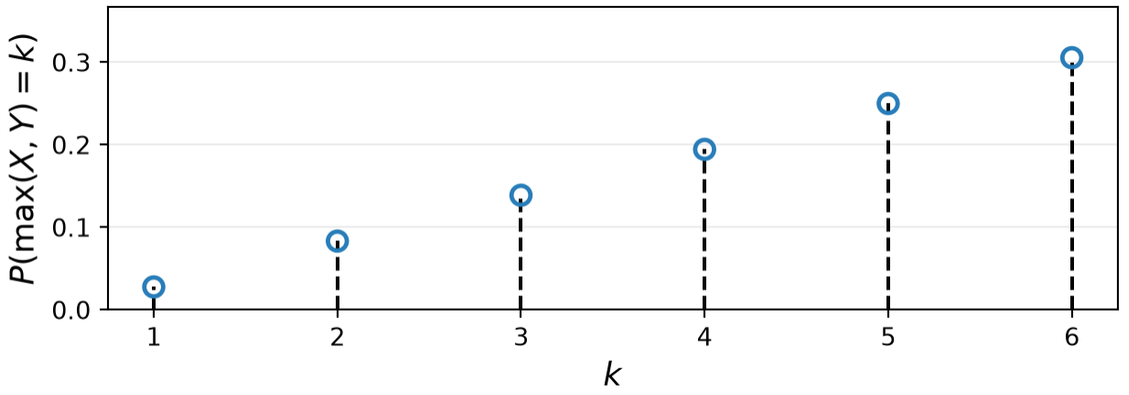
Example: Maximum of Two Dice Rolls

Example 3.7.6: We roll two fair 6-sided dice. Let X be the number on the first die and Y the number on the second. What is the PMF of $\max(X, Y)$?

Every one of the 36 outcomes is equally likely, so we count how many land on each value of the maximum:

$$P\left(\max(X, Y) = k\right) = \frac{2k - 1}{36}$$





Part III: Independence of Random Variables

Independence of Random Variables

Random variables X and Y are independent if

$$P(X \leq x,\ Y \leq y) = P(X \leq x)\, P(Y \leq y)$$

for all $x, y \in \mathbb{R}$.

In the discrete case, where we are now working, this is equivalent to

$$P(X = x,\ Y = y) = P(X = x)\, P(Y = y)$$

ⓘ **If X and Y are independent R.V.s, then *any* function of X is independent of *any* function of Y.**

Independent and Identically Distributed

We will often work with independent and identically distributed (i.i.d.) random variables.

- **Independent**: they provide no information about each other
- **Identically distributed**: they come from the same exact PMF

Give an example of two variables that are:

1. Independent and identically distributed
2. Independent and *not* identically distributed
3. Dependent and identically distributed
4. Dependent and *not* identically distributed

Conditional Independence of Random Variables

Random variables X and Y are conditionally independent given a R.V. Z if

$$P(X \leq x,\ Y \leq y \mid Z = z) = P(X \leq x \mid Z = z)\, P(Y \leq y \mid Z = z)$$

Same definition as before - every probability in it is now conditional on $Z = z$.

Independence and Conditional Independence

Independence does *not* imply conditional independence and conditional independence does *not* imply independence.

Role of Z	Structure	Example	Before conditioning	After conditioning	Effect of conditioning
Confounder	$X \leftarrow Z \rightarrow Y$	Smoking (Z) makes people both drink more coffee (X) and get lung cancer (Y). Among people with the same smoking habits, coffee and cancer are unrelated.	dependent	independent	removes a spurious association
Mediator	$X \rightarrow Z \rightarrow Y$	A drug (X) prevents strokes (Y) by lowering blood pressure (Z). Among patients with the same blood pressure, the drug appears useless.	dependent	independent	removes the entire effect
Collider	$X \rightarrow Z \leftarrow Y$	Two unrelated diseases (X and Y) can each put someone in the hospital (Z). Among hospitalized patients, the diseases appear negatively related: a patient who doesn't have one probably has the other.	independent	dependent	creates a spurious association

Part IV: Expectation

Definition of Expectation

Usually we want to *summarise* a R.V. rather than write out all its values and their probabilities.

The expected value of a random variable (also called the expectation or the mean) is the weighted average of the possible values that X can take on, weighted by their probabilities:

$$E(X) = \sum_{x} \underbrace{x}_{\text{value}} \underbrace{P(X = x)}_{\text{PMF at } x}$$

where the sum is over all possible values of x.

Examples of Expectations

1. Let X be the result of rolling a 6-sided die. Then

$$E(X) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

ⓘ X never *equals* its mean, which is fine. The average number of children per household in some country could be 1.8, but no household has exactly 1.8 children.

2. Let $X \sim \text{Bern}(p)$. Then

$$E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$$

Linearity of Expectation

A handy property of expectation is linearity:

$$E(cX) = c\,E(X)$$
$$E(X+Y) = E(X) + E(Y)$$

> ⓘ **Linearity holds whether or not X and Y are independent!**

Consider the most extreme dependence, $X = Y$:

Then $X + Y = 2X$

$E(X+Y) = E(2X) = 2E(X)$

$E(X) + E(Y) = 2E(X)$ ✓

Recall that the R.V. X assigns a real number $X(s)$ to each possible outcome of the sample space S.

$$E(X+Y) = \sum_{s\in S} (X(s) + Y(s))\,P(s) = \sum_{s\in S} X(s)P(s) + \sum_{s\in S} Y(s)P(s) = E(X) + E(Y)$$

Linearity Stops at Linear Functions

But this only works for *linear* functions. In general $E\big(g(X)\big)$ is **not equal to** $g\big(E(X)\big)$ when g is nonlinear:

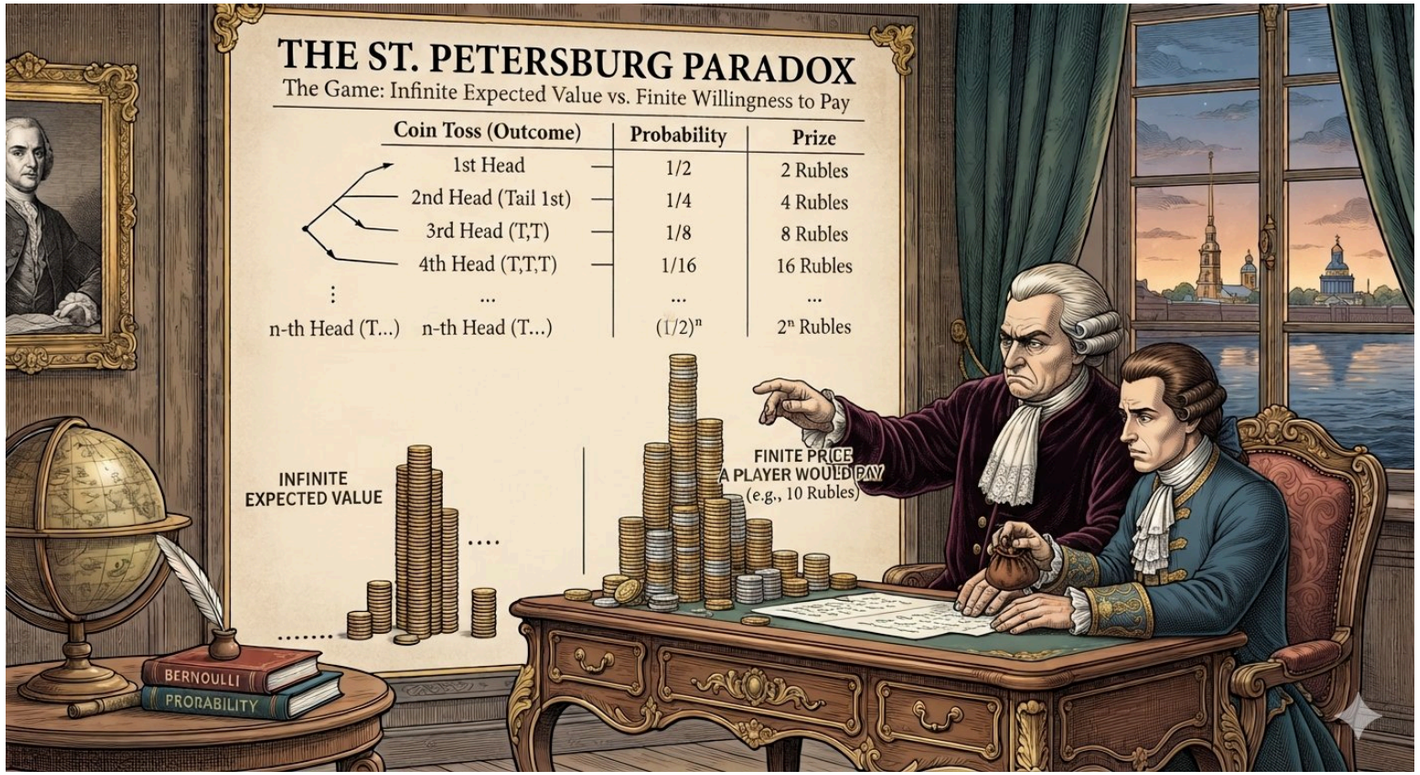
- $g(x) = x^3$
- $g(x) = \sqrt{x}$
- $g(x) = |x|$

In all of these cases, you have to push the function through the PMF. This is the law of the unconscious statistician (LOTUS):

$$E\big(g(X)\big) = \sum_x g(x)\, P(X = x)$$

> ⓘ **The average of a function is not the function of the average**

The St. Petersburg Paradox



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The St. Petersburg Paradox

Expected value

$$E(X) = \sum_{k}^{\infty} 2^k P(k) = \sum_{k}^{\infty} 2^k \left(\frac{1}{2}\right)^k = \sum_{k}^{\infty} 1 = \infty$$

Finite Lotery $E_L(X) = L$

	Bankrol	$E(X)$
Millionaire	1,050,000	20
Billionaire	1,075,000,000	30
World GDP (2020)	83.8 trillion	46
Atoms in the universe	10^{80}	266
Googolionaire	100^{100}	332

See https://en.wikipedia.org/wiki/St._Petersburg_paradox

For more on large numbers checkout From 1,000,000 to Graham's Number blog post on waitbutwhy.com .

Variance

The variance of a R.V. X is defined as

$$\text{Var}(X) = E\left((X - EX)^2\right)$$

and the standard deviation is the square root of the variance.

Intuitively, this is the average (squared) deviation of the variable from its mean. *(Why squared?)*

Example: What is the variance of one roll of a fair six-sided die? Recall $E(X) = 3.5$:

$$\frac{1}{6}\left((1 - 3.5)^2 + (2 - 3.5)^2 + \cdots + (6 - 3.5)^2\right) \approx 2.9$$

and the standard deviation is about 1.7.

Another Useful Definition of Variance

Another way to compute the variance is

$$\mathrm{Var}(X) = E(X^2) - \Big(E(X)\Big)^2$$

Let's prove this from the definition, letting $E(X) = \mu$.

Note that $E\Big((X-\mu)^2\Big) \neq \Big(E(X-\mu)\Big)^2$, since this function of X is nonlinear.

Instead:

$$\begin{aligned} E\Big((X-\mu)^2\Big) &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - E(2\mu X) + E(\mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \end{aligned}$$

Some Useful Variance Facts

1. $\mathrm{Var}(c) = 0$ for any constant c
2. $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ for any constant c
3. $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$ for any constant c
4. $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ only if X and Y are independent (not in general)

Unlike expectation, variance is *not* linear

In general $\mathrm{Var}(cX) \neq c\,\mathrm{Var}(X)$ and $\mathrm{Var}(X + Y) \neq \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Part V: Joint Distributions

Introducing Joint Distributions

- We have learned that two random variables can be independent or dependent
- We have learned how to characterise PMFs and CDFs of functions of discrete random variables
- But what if we want more detailed information about how two variables interact?

Examples abound: effectiveness of a medical treatment, interactions between genetic markers of a disease, electoral conditions.

The Joint PMF and CDF

The joint PMF of discrete variables X and Y is defined as

$$P_{X,Y}(x, y) = P(X = x,\ Y = y)$$

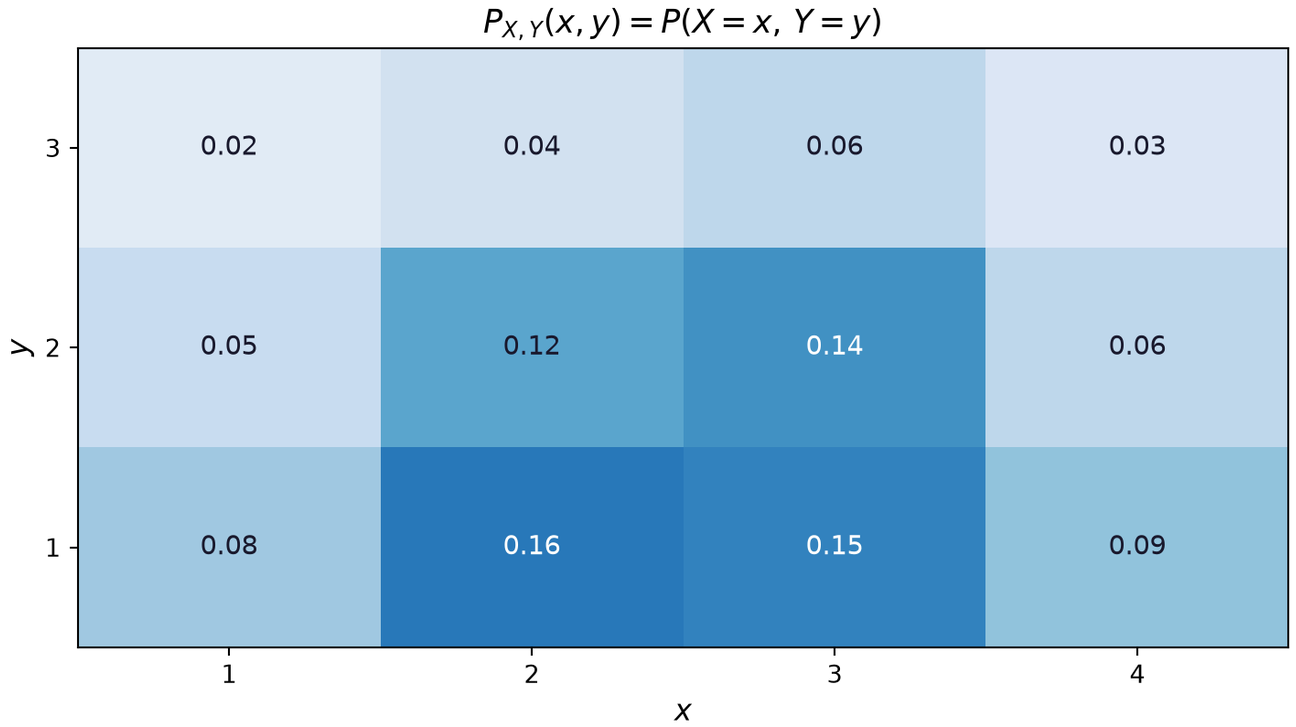
Much like a univariate PMF, a joint PMF has to sum to 1 over all values of X and Y:

$$\sum_x \sum_y P(X = x,\ Y = y) = 1$$

The joint CDF is defined as

$$F_{X,Y}(x, y) = P(X \leq x,\ Y \leq y)$$

A Joint PMF



Each cell is one (x, y) pair; the numbers sum to 1 over the whole table.

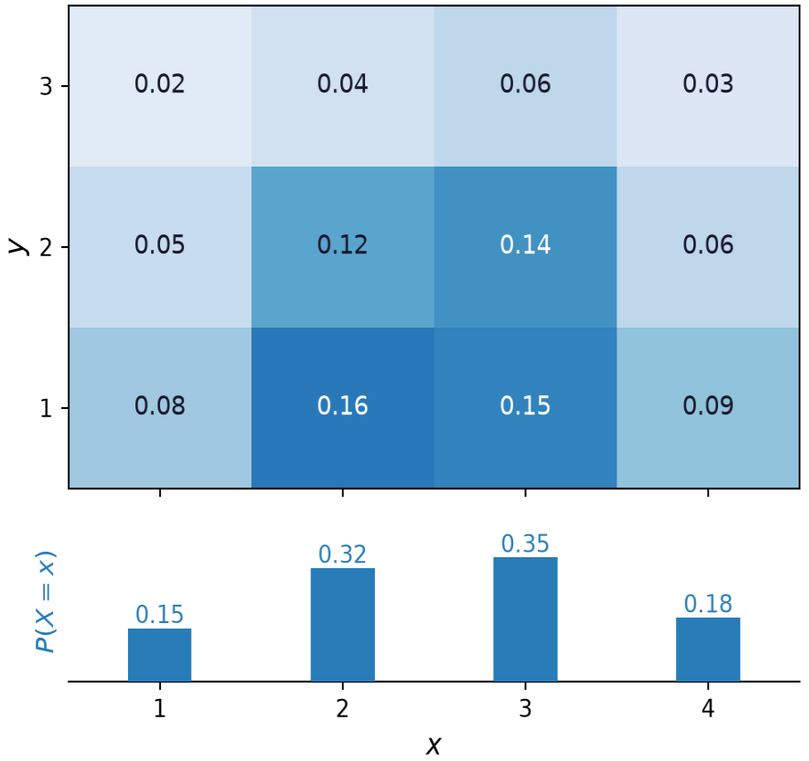
The Marginal PMF

The marginal PMF of X is defined as

$$P(X = x) = \sum_y P(X = x,\ Y = y)$$

The marginal PMF of the joint distribution of X and Y *is* the PMF of X.

The act of summing over the possible values of Y to get the PMF of X is called marginalizing out Y.



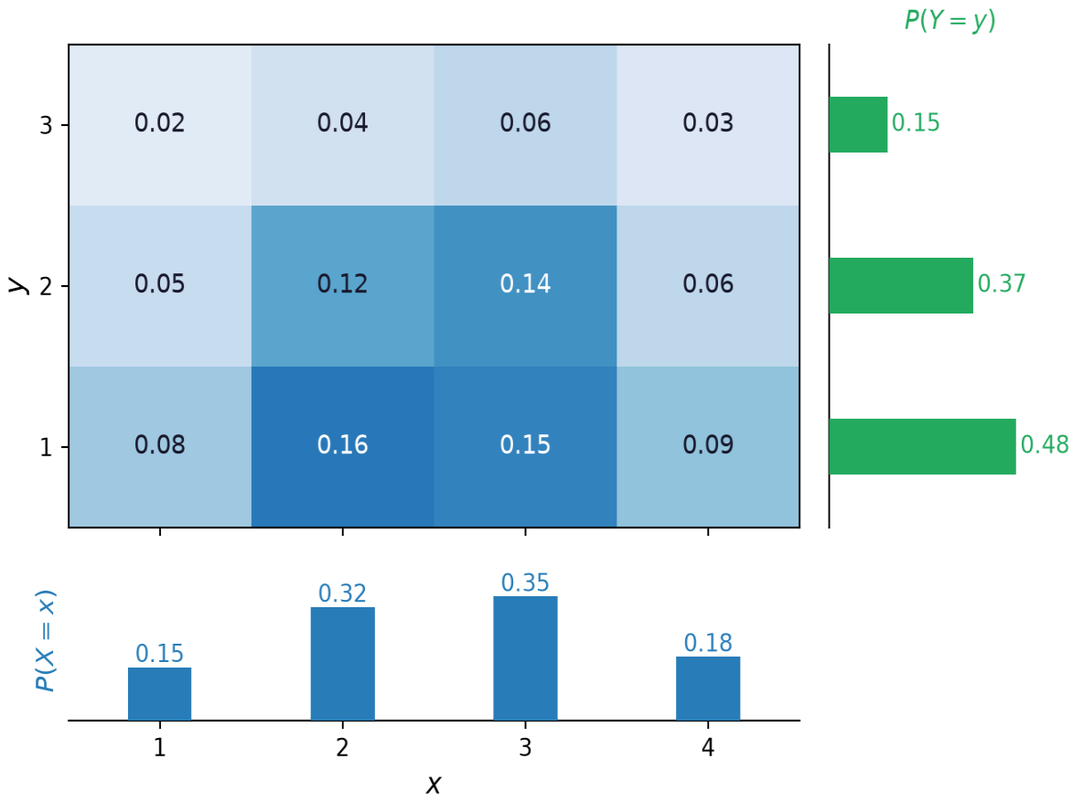
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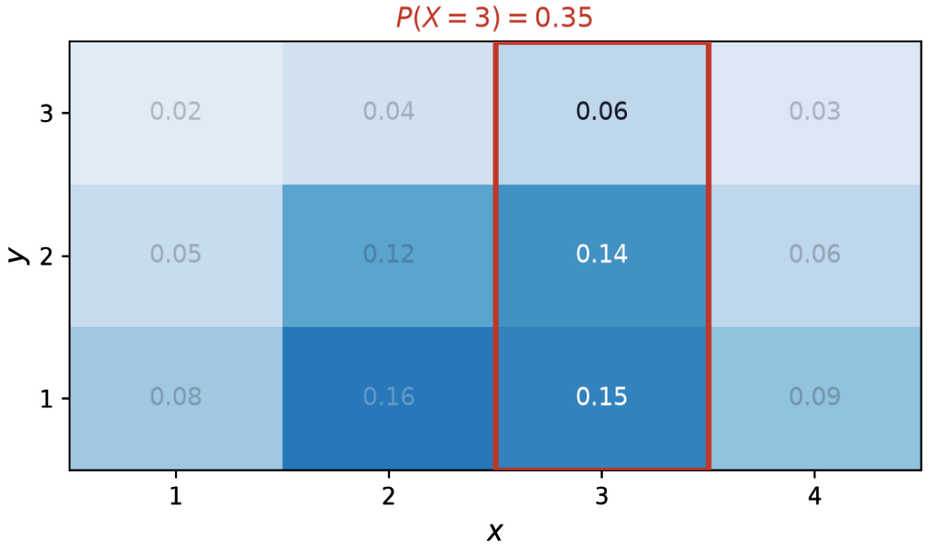
Sum down a column for $P(X = x)$; sum across a row for $P(Y = y)$.

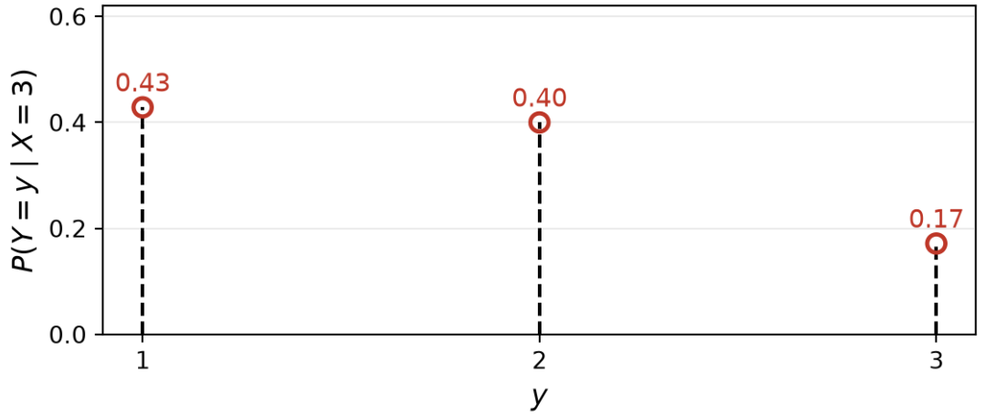
The Conditional PMF

The conditional PMF of Y given $X = x$ is

$$P(Y = y \mid X = x) = \frac{P(X = x,\ Y = y)}{P(X = x)}$$

Pick one column of the joint table and rescale it so that it sums to 1 again.





The highlighted column, divided by its own total - a valid PMF in its own right.

Putting It All Together

The following contingency table illustrates the above concepts.

Suppose we randomly sample an adult male from the population. Let X indicate the presence of a gene and Y indicate developing a certain disease at some point in his life.

	$Y = 1$	$Y = 0$
$X = 1$	$\frac{5}{100}$	$\frac{20}{100}$
$X = 0$	$\frac{3}{100}$	$\frac{72}{100}$

- What is the joint PMF of X and Y?

The contingency table itself - it already lists $P(X = x,\ Y = y)$ for all four cells.

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Suppose we randomly sample an adult male from the population. Let X indicate the presence of a gene and Y indicate developing a certain disease at some point in his life.

	$Y = 1$	$Y = 0$	**Total**
$X = 1$	$\frac{5}{100}$	$\frac{20}{100}$	$\frac{25}{100}$
$X = 0$	$\frac{3}{100}$	$\frac{72}{100}$	$\frac{75}{100}$
Total	$\frac{8}{100}$	$\frac{92}{100}$	$\frac{100}{100}$

- What is the joint PMF of X and Y?

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- What are the marginal probabilities of X and Y?

The row and column totals: $X \sim \text{Bern}(0.25)$ and $Y \sim \text{Bern}(0.08)$.

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- What is the conditional probability of Y given $X = 1$?

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1,\ Y = 1)}{P(X = 1)} = \frac{5/100}{25/100} = 0.2$$

So Y given $X = 1$ is $\text{Bern}(0.2)$.

Independence, Revisited

Random variables X and Y are independent if

$$P(X = x,\ Y = y) = P(X = x)\,P(Y = y)$$

- that is, if the joint table is exactly the outer product of its own margins.

Are These Two Variables Independent?

	$Y = 1$	$Y = 0$	**Total**
$X = 1$	$\frac{5}{100}$	$\frac{20}{100}$	$\frac{25}{100}$
$X = 0$	$\frac{3}{100}$	$\frac{72}{100}$	$\frac{75}{100}$
Total	$\frac{8}{100}$	$\frac{92}{100}$	$\frac{100}{100}$

$$P(X = 1,\ Y = 1) = 0.05 \qquad \text{but} \qquad P(X = 1)\,P(Y = 1) = 0.25 \times 0.08 = 0.02$$

Finding even **one** pair of values such that $P(X = x, Y = y) \neq P(X = x)P(Y = y)$ is enough to rule out independence.

Summary

Concept	Key formula
Function of two r.v.s	$g(X, Y)$ maps $s \mapsto g(X(s), Y(s))$
Independence of r.v.s	$P(X = x, Y = y) = P(X = x)P(Y = y)$
Conditional independence	$P(X \leq x, Y \leq y \mid Z = z) = P(X \leq x \mid Z = z)P(Y \leq y \mid Z = z)$
Expectation	$E(X) = \sum_x x\, P(X = x)$
Linearity	$E(cX + Y) = c\, E(X) + E(Y)$, always
Nonlinear g (LOTUS)	$E(g(X)) = \sum_x g(x)P(X = x) \neq g(E(X))$
Variance	$\mathrm{Var}(X) = E((X - EX)^2) = E(X^2) - (EX)^2$
Joint PMF	$p_{X,Y}(x, y) = P(X = x, Y = y)$
Marginal PMF	$P(X = x) = \sum_y P(X = x, Y = y)$
Conditional PMF	$P(Y = y \mid X = x) = p_{X,Y}(x, y)/P(X = x)$